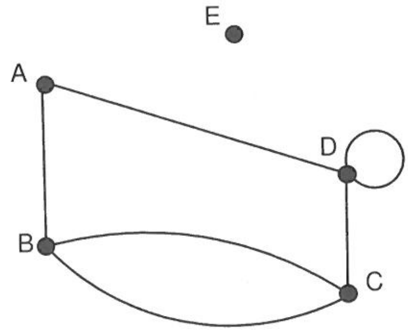
Adjacency Matrices

Adjacent vertices

Two vertices are said to be **adjacent** if they are connected by an edge.



Consider the graph on the right.
Note that two vertices may be connected by one edge, more than one edge (B and C are connected by 2 distinct edges), that a vertex may be connected to itself (vertex D), and that a vertex need not be connected to any other vertex (vertex E).

Undirected Graphs

Adjacency Matrix

An adjacency matrix for an **undirected** graph containing *n* vertices is a square matrix of order *n* x *n* in which the entry in row *i* and column *j* is the number of edges joining the vertices *i* and *j*.
Note that in an adjacency matrix a loop is counted as 1 edge.
For the given graph, which is undirected, the adjacency matrix is:

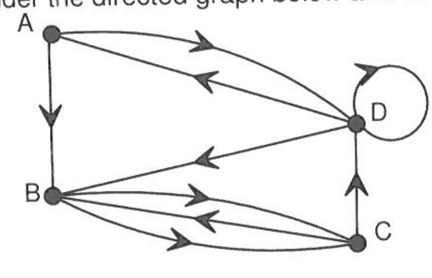
$$\begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array}\begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Matrix Properties of an Undirected Graph

- an adjacency matrix is a square matrix of order n x n where n is the number of graph vertices,
- the entry in row *i*, column *j* is the **number of edges** there are joining vertex *i* to vertex *j*,
- the entries in the matrix are **symmetrical across the leading diagonal**, that is any entry in row *i* and column *j* is the same as the entry in row *j* and column *i*,
- if the graph has no loops then the leading diagonal of the matrix shows only zeros, in this case the graph has a loop at vertex D, hence there is an entry of 1 in the leading diagonal,
- a loop is counted as one edge,
- the sum of the row entries gives the degree of the vertex corresponding to that row. For example, row 3 of the adjacency matrix corresponds to vertex C of the graph and the sum of the entries in row 3 is 0 + 2 + 0 + 1 + 0 = 3 which is the degree of vertex C. For any vertex containing a loop one must be added because in this context a loop is one edge but contributes 2 to the degree of a vertex, (Check that you agree that the degree of vertex D is 4.)
- if the graph has no multiple edges then all entries are either 1 or 0. In the given graph there are multiple edges between B and C, hence the entry 2 in the adjacency matrix.

Directed Graphs

Consider the directed graph below and its adjacency matrix shown on the right.



$$\begin{array}{c} \\ A \\ B \\ C \\ D \end{array}\begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

Adjacency Matrix

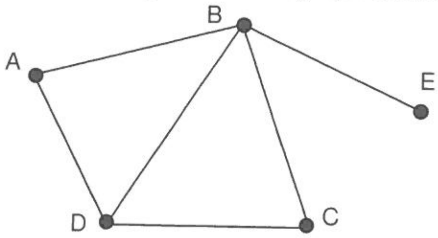
An adjacency matrix for a **directed** graph containing *n* vertices is a square matrix of order *n* x *n* in which **the entry in row *i* and column *j* is the number of directed edges** joining the vertex *i* and *j* in the direction *i* to *j*.

Matrix Properties of a directed Graph

- an adjacency matrix is a square matrix of order n x n where n is the number of graph vertices,
- the entry in row *i*, column *j* is the **number of directed edges** there are joining vertex *i* to vertex *j*,
- the entries in the matrix are **not symmetrical across the leading diagonal**,
- if the graph has no loops then the leading diagonal of the matrix shows only zeros, in this case the graph has a loop at vertex D, hence there is an entry of 1 in the leading diagonal,
- the sum of the entries in a column is equal to the in-degree of the vertex that corresponds to that column,
- the sum of the entries in a row is equal to the out-degree of the vertex that corresponds to that row.

Powers of the Adjacency Matrix - Undirected Graphs

Consider the following undirected graph and its adjacency matrix, M, shown below:



$$M = \begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

The graph and the matrix, M, both show that there:

- are 0 walks of length 1 starting at A and finishing at A.
- is 1 walk of length 1 starting at B and finishing at D.
- is 1 walk of length 1 starting at B and finishing at E.

If we consider walks of length 2 the graph shows the following:

- there are 2 walks of length 2 starting at A and finishing at A, these walks are ABA and ADA.
- there is only 1 walk of length 2 starting at A and finishing at B, this walk is ADB.
- there are 2 walks of length 2 starting at A and finishing at C, these walks are ABC and ADC.
- there are 4 walks of length 2 starting at B and finishing at B, these walks are BAB, BDB, BCB and BEB.
- there are 0 walks of length 2 starting at B and finishing at E.

If you recall your study of matrices Unit One Applications, then the number of walks of length 2 between the vertices of a graph can be obtained from the square of the adjacency matrix.
The matrix M^2 is given below on the left.

$$M^2 = \begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 2 & 1 & 2 & 1 & 1 \\ 1 & 4 & 1 & 2 & 0 \\ 2 & 1 & 2 & 1 & 1 \\ 1 & 2 & 1 & 3 & 1 \\ 1 & 0 & 1 & 1 & 1 \end{bmatrix} \end{array} \qquad M^3 = \begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 2 & 6 & 2 & 5 & 1 \\ 6 & 4 & 6 & 6 & 4 \\ 2 & 6 & 2 & 5 & 1 \\ 5 & 6 & 5 & 4 & 2 \\ 1 & 4 & 1 & 2 & 0 \end{bmatrix} \end{array}$$

The matrix M^3 gives the number of walks of length 3 between selected vertices.
The entry in row 1 column 1 in matrix M^3 is 2.
The matrix informs us that there are 2 walks of length 3 starting at A and finishing at A.
To determine these two walks we need to refer back to the graph.
The two walks under discussion are ABDA and ADBA.
Note that there are: 3 edges in each of these walks (For walk ABDA edges are AB, BD, and DA), and
2 stop-overs (For walk ABDA stop-overs are the vertices B and D).

Matrix M^3 informs us that there are 5 walks of length 3 starting at C and finishing at D. These walks are CBAD, CBCD, CDCD, CDBD and CDAD.

Consider matrix N where $N = M + M^2$.

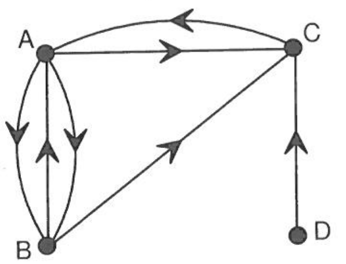
$$N = \begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \end{array} + \begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 2 & 1 & 2 & 1 & 1 \\ 1 & 4 & 1 & 2 & 0 \\ 2 & 1 & 2 & 1 & 1 \\ 1 & 2 & 1 & 3 & 1 \\ 1 & 0 & 1 & 1 & 1 \end{bmatrix} \end{array} = \begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 2 & 2 & 2 & 2 & 1 \\ 2 & 4 & 2 & 3 & 1 \\ 2 & 2 & 2 & 2 & 1 \\ 2 & 3 & 2 & 3 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

Matrix M gives the number of walks of length 1 (i.e. 1 step) between selected vertices (no stop-overs).
Matrix M^2 gives the number of walks of length 2 (i.e. 2 steps) between selected vertices (one stop-over).
Matrix N, where $N = M + M^2$, gives the total number of walks of length 1 or length 2 between selected vertices.
Alternatively we could say that matrix N gives the number of walks with at most one stop-over.

The matrix $M + M^2 + M^3$ gives the number of walks with at most two stop-overs etc.

Powers of the Adjacency Matrix - Directed Graphs

Consider the following directed graph (digraph) and its adjacency matrix, P, shown below:



$$P = \begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 2 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

The graph and the matrix, M, both show that there:

- are 2 walks of length 1 starting at A and finishing at B.
- is 1 walk of length 1 starting at A and finishing at C.
- is 1 walk of length 1 starting at B and finishing at A.

If we consider walks of length 2 the graph shows the following:

- there are 3 walks of length 2 starting at A and finishing at A, these walks are ABA, ABA and ACA.
- there are 2 walks of length 2 starting at A and finishing at C, these walks are ABC and ABC.
- there is only 1 walk of length 2 starting at B and finishing at C, this walk is BAC.
- there is only 1 walk of length 2 starting at D and finishing at A, this walk is DCA.

The matrix P^2 below on the left shows the number of walks of length 2 between selected vertices.

$$P^2 = \begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 3 & 0 & 2 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{array} \qquad P^3 = \begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 2 & 6 & 3 & 0 \\ 3 & 2 & 3 & 0 \\ 3 & 0 & 2 & 0 \\ 0 & 2 & 1 & 0 \end{bmatrix} \end{array}$$

The matrix P^3 gives the number of walks of length 3 between selected vertices.

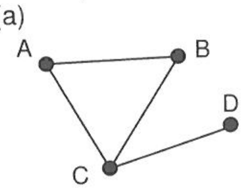
NOTE: If A is an adjacency matrix for a given **undirected or directed** graph, then

(i) A is a one stage matrix with no stop-overs,
(ii) A^2 gives the two stage matrix with one stop-over,
(iii) A^3 gives the three stage matrix with two stop-overs, etc.
(iv) $A + A^2$ gives a matrix with **at most** one stop-over,
(iv) $A + A^2 + A^3$ gives a matrix with **at most** two stop-overs, etc.

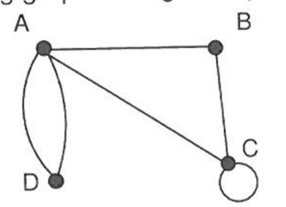
EXERCISE 7E

1. Represent each of the following graphs using an adjacency matrix.

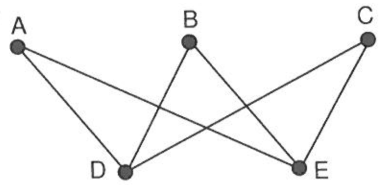
(a)



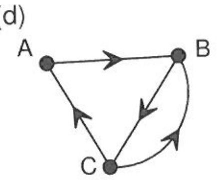
(b)



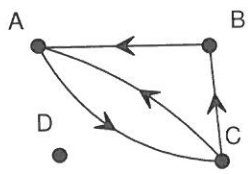
(c)



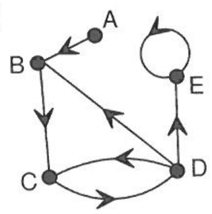
(d)



(e)



(f)



2. Use the following adjacency matrices to draw a corresponding planar graph.

(a)
$$\begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

(b)
$$\begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 2 & 1 & 0 \\ 2 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

(c)
$$\begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 0 \\ 3 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

(d)
$$\begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 2 & 0 & 1 \\ 0 & 2 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

(e)
$$\begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

(f)
$$\begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

3. (a) (i) Calculate the degree sum for your graph in 2.(a).

(ii) Calculate the sum of the entries in the adjacency matrix in 2.(a).

(iii) Comment on your answers to (i) and (ii) above.

(b) (i) Calculate the degree sum for your graph in 2.(b).

(ii) Calculate the sum of the entries in the adjacency matrix in 2.(b).

(iii) Comment on your answers to (i) and (ii) above.

(c) Why are your answers to (a) (iii) and (b) (iii) different?

(d) (i) Calculate the in-degree sum for your graph in 2.(e).

(ii) Calculate the out-degree sum for your graph in 2.(e).

(iii) Calculate the sum of the entries in the adjacency matrix in 2.(e).

(iv) Comment on your answers.

(e) How will a directed loop added to one of the vertices in graph 2.(e) affect your answers to (d) above?

4. Consider the following adjacency matrices.

$$\begin{array}{c} \\ A \\ B \\ C \\ D \end{array}\begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 3 & 1 & 0 \\ 3 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

Matrix P

$$\begin{array}{c} \\ A \\ B \\ C \\ D \end{array}\begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

Matrix Q

$$\begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array}\begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Matrix R

$$\begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array}\begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Matrix S

$$\begin{array}{c} \\ A \\ B \\ C \\ D \end{array}\begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Matrix T

Match each matrix with one of the following graph types/descriptions. State a feature of the matrix that helped you to decide on the graph type.

(a) Digraph:

(b) Bipartite graph:

(c) Complete graph:

(d) Graph with a loop:

(e) Disconnected graph:

5. Present each of the matrices in question 4 above as a graph.

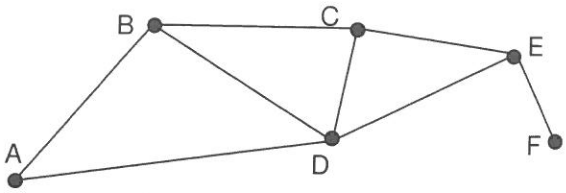
6. The adjacency matrices, A and B, for A-Air Airlines and B-Air Airlines between the cities A, B, C and D are given below. A "1" indicates a direct flight exists between the given cities and a "0" indicates no direct flight exists between the given cities.

$$A = \begin{array}{c} \\ A \\ B \\ C \\ D \end{array}\begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array} \qquad B = \begin{array}{c} \\ A \\ B \\ C \\ D \end{array}\begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

(a) Examine each matrix and state which airline has flights which are "one way only"? Justify your answer.

(b) Verify your answer to (a) by constructing a graph of the flight routes to both airlines.

7. The diagram on the right shows the direct rail links between six towns A, B, C, D, E and F.



(a) How many two-step routes can a traveler take in travelling from A to C? List them.

(b) How many routes with just one stop-over can a traveler take in travelling from B to B? List them.

(c) How many three-step routes can a traveler take in travelling from A to C? List them.

(d) How many routes with two stop-overs can a traveler take in travelling from E to A? List them.

(e) How many three-step routes can a traveler take in travelling from F to F? List them.

(f) Write down the adjacency matrix, A, for the graph. Explain what it means.

(g) Use your calculator to find A^2. What does this matrix tell you?

(h) Use your calculator to verify A^3 given below. Explain what it tells you.

$$A^3 = \begin{array}{c} \\ A \\ B \\ C \\ D \\ E \\ F \end{array} \begin{array}{c} \begin{array}{cccccc} A & B & C & D & E & F \end{array} \\ \begin{bmatrix} 2 & 5 & 3 & 6 & 3 & 1 \\ 5 & 4 & 7 & 7 & 3 & 2 \\ 3 & 7 & 4 & 7 & 6 & 1 \\ 6 & 7 & 7 & 6 & 7 & 1 \\ 3 & 3 & 6 & 7 & 2 & 3 \\ 1 & 2 & 1 & 1 & 3 & 0 \end{bmatrix} \end{array}$$

(i) What use can be made of A^3 in this question?

(j) What does the zero in A^3 tell you?

(k) The entry a^3_{25} is 3. What does this mean?

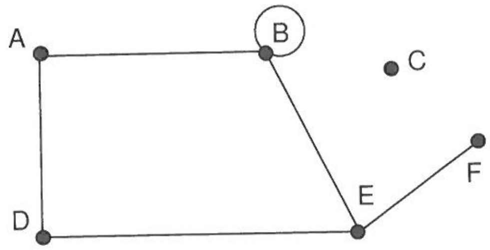
(l) Determine matrix S where $S = A + A^2$. What does S tell you?

(m) What is the s_{16}? What does it tell you?

(n) Determine matrix T where $T = A + A^2 + A^3$. What does T tell you?

(o) Matrix T does not contain any zero entries. What does this tell you?

8. Consider the undirected graph on the right.
The graph shows that there :
- is 1 one-step sequence of edges from A to D
- is 1 one-step sequence from B to B
- are 2 two-step sequences from A to E, they are ABE and ADE
- are 3 two-step sequences from F to E, they are FEBE, FEDE and FEFE



(a) List all the two-step sequences from B to D.

(b) List all the two-step sequences from E to E.

(c) List all the three-step sequences from A to F.

(d) List all the three-step sequences from A to D.

(e) Construct, M, the adjacency matrix for the given graph.

(f) Determine M^3, and write it below.

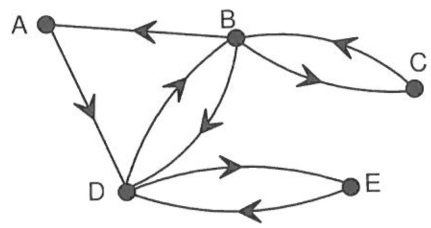
(g) Matrix M^3 shows that $m_{55}^3 = 1$.
What does this entry tell you?

(h) List all the sequences identified by m_{25}^3.

(i) Classify the graph under consideration.

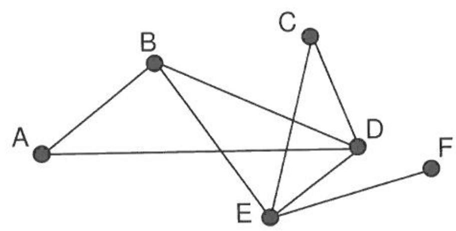
(j) In the adjacency matrix M, what do row 3 and column 3 have in common? Why?

9. Consider the digraph shown on the right.
(a) **Explain** how you would go about finding the number of three-step walks from B to D.



(b) List all of the three-step walks from B to D.

10. The graph of the right shows the route map for Q-air Airlines.
A, B, C, D, E and F represent cities on the route map.
(a) Determine Q, the adjacency matrix for this graph.



Q-Air Airlines claims that it is possible to fly with them between any of these six cities, in at most, two flights, that is, one stop-over. Mathematically, justify or reject, Q-Air's claim.